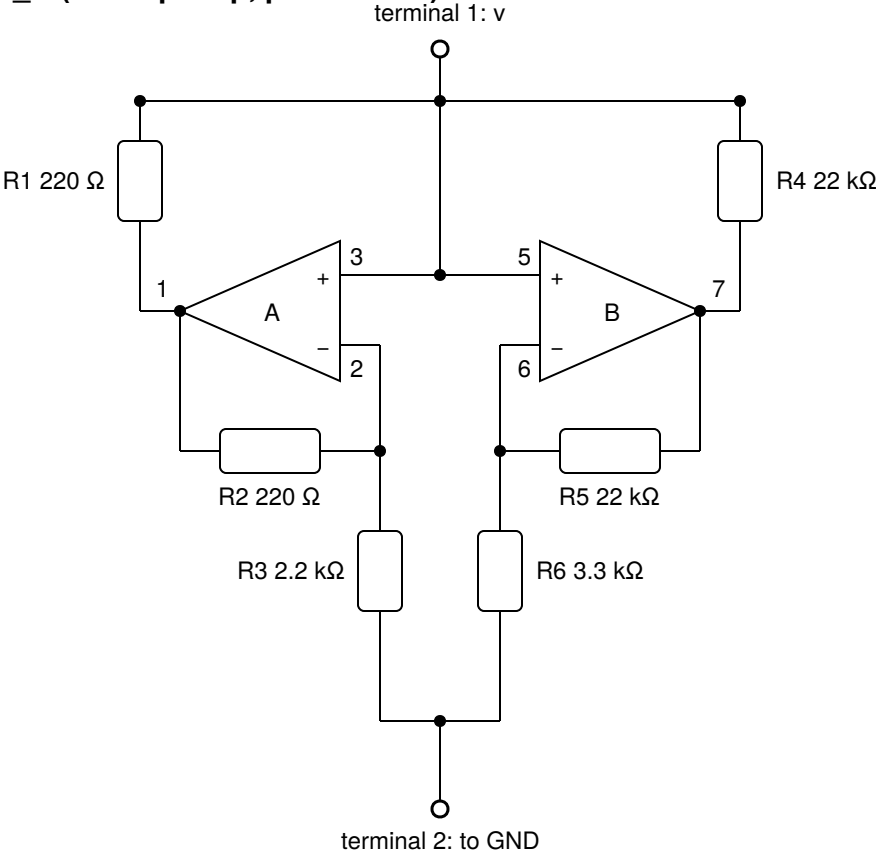
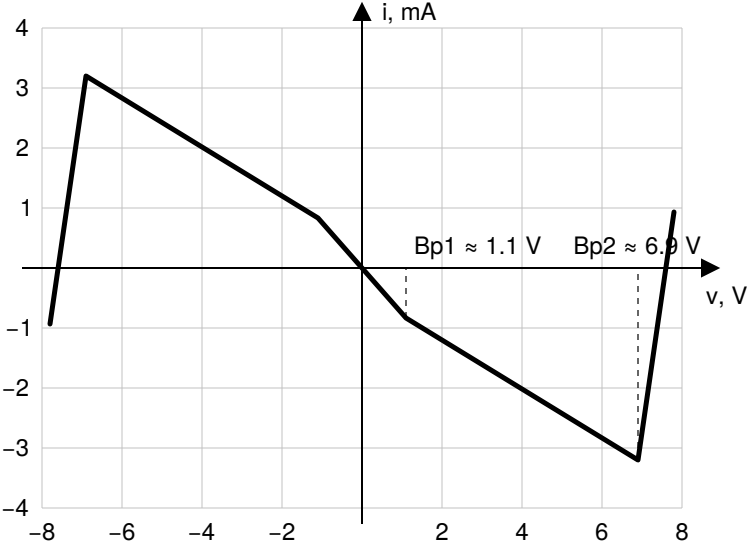


Chua diode N_R (dual op-amp, pins shown)



Expected V-I characteristic $i(v)$



$R_a = -(R_3 \parallel R_6) \approx -1.32$ k Ω (inner segment, $|v| < Bp1$)
 $R_b = -R_3 R_4 / (R_4 - R_3) \approx -2.44$ k Ω (middle segments)
outer segments (op-amp saturation): $\approx +R_1 \parallel R_4 \approx +220$ Ω

Supply decoupling (TL082, shared pins 8 / 4)

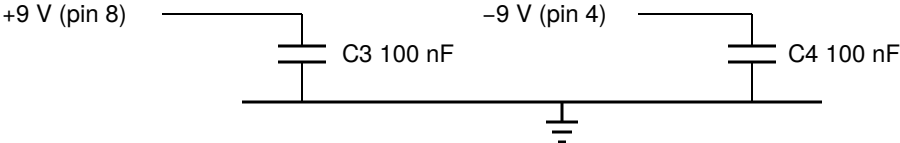
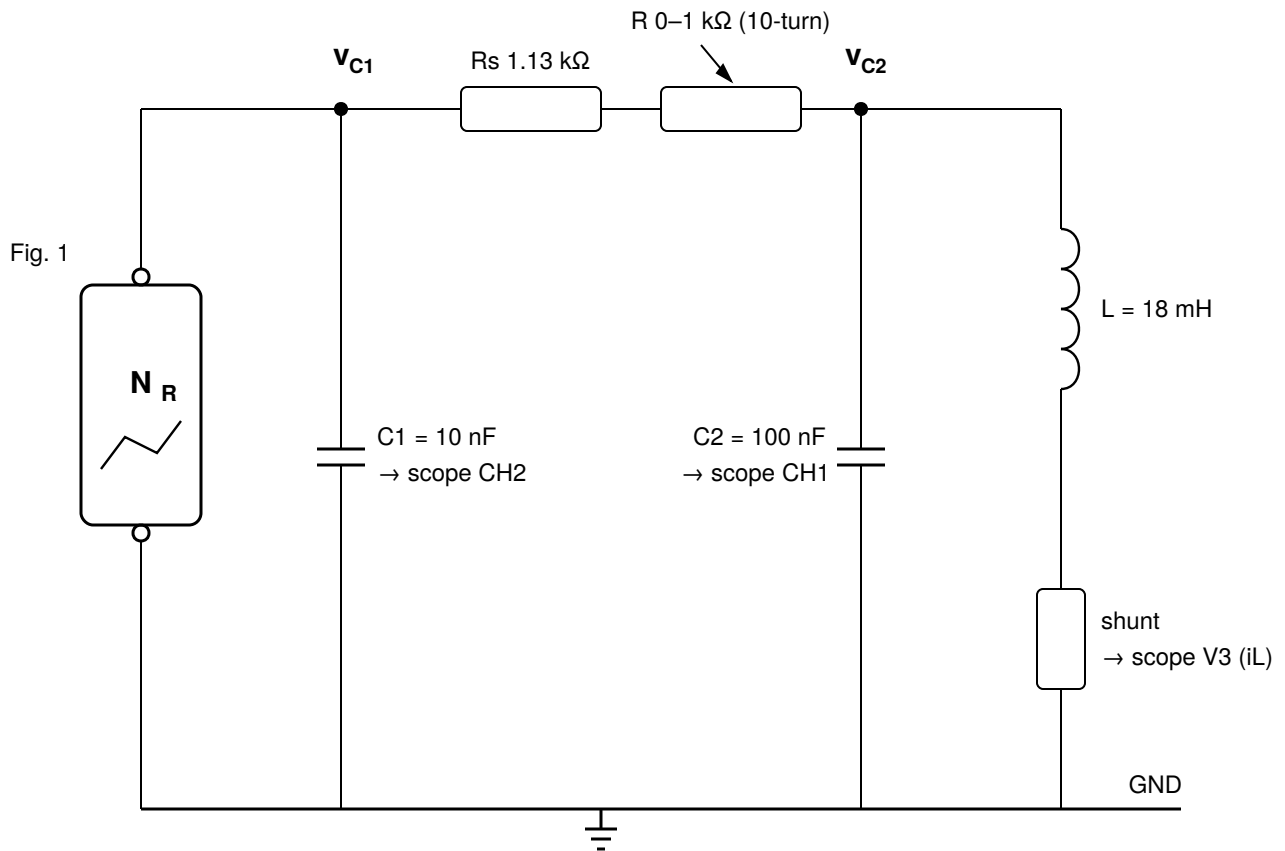


Fig. 1. Chua diode as a two-terminal element: circuit, expected piecewise-linear V-I curve, and supply decoupling.

Chua oscillator (N_R shown as a two-terminal element)



Total coupling resistance $R_{\text{total}} = R_s + R = 1.13 \dots 2.13 \text{ k}\Omega$; measurement points: 1.54 / 1.74 / 1.91 k Ω .

Dimensionless parameters: $\alpha = C2/C1 = 10$, $\beta = R_{\text{total}}^2 \cdot C2/L = 13 \dots 20$.

Fig. 2. Chua oscillator: linear resonant part (L , $C2$), coupling resistance, $C1$, and the nonlinear element N_R of Fig. 1.